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Q1)

```
product="ice water"
flavour="strawberry"
quantity=8

if product == "ice cream":
    if flavour != "red bean":
        if quantity < 4:
            print("2 dollars")
        elif quantity > 20:
            print("17 dollars")
elif product == "ice lolly":
    if flavour == "strawberry":
        if quantity > 20:
            print("10 dollars")
        elif quantity != 0:
            print("4 dollars")
        else:
            print("0 dollars")
    else:
        print("3 dollars")
elif quantity < 8:
    print("9 dollars")
else:
    if quantity == 8:
        if flavour == "chocolate":
            print("20 dollars")
        elif quantity != 8:
            if product == "ice water":
                print("15 dollars")
            elif product == "ice cream":
                print("16 dollars)
        else:
            print("5 dollars")
    else:
        if product == "ice water":
            if flavour != "vanilla":
                print("7 dollars")
        else:
            if flavour != "chocolate":
                if quantity == 18:
                    print("12 dollars")
```

What does the program print when it is executed?

Answer: The program prints _____

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Q2)
Here is a description of the Hong Kong 'Mark 6' lottery:

> *"Mark Six is a lottery style game where players guess what numbers will be drawn out of a lottery machine. The lottery machine is a transparent plastic cylinder which contains colored balls numbered 1 to 49. The cylinder is rotated to provide randomization in the drawing process. Players choose six different numbers out of a possible 49."*

Somebody has created a Python program to generate a set of 6 numbers which can be used for the lottery. The program is shown below.

```
import random

print("This is the Mark 6 random number generator.")
print("I will generate 6 random numbers for you.")

for i in range(6):
    print(random.randint(1, __________ ) )  # Write your answer below, not here

print("Finished! Good luck with these numbers!")
```

(It is possible that the same number is generated more than once, but that is irrelevant for this exam question). Question: what **number** must be written in the area shown as ____ above ?
You cannot alter any of the code given or add any new code.

Answer: The number is _____

Q3)
Two friends take turns paying for their meals each day.

```
day_number=1
total_cost=0
this_day_payer="friend1" # First person to pay

while day_number<6:
    if day_number == 3:
        print("This is day", day_number, "- a birthday party, paid by both!")
        total_cost=total_cost+400
    elif this_day_payer == "friend1":
        print("This is day", day_number, "- friend1 buys cheap food")
        total_cost=total_cost+100
        this_day_payer="friend2"
    elif this_day_payer == "friend2":
        print("This is day", day_number, "- friend2 buys good food")
        total_cost=total_cost+200
        this_day_payer="friend1"
    day_number=day_number+1

print("Total amount spent on meals =", total_cost)
```

No error occurs when the program is run. According to the last message shown by the program, what is the **total amount** spent on meals? Write the number clearly in the space below.

Answer: The **total amount** spent on meals is _____

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Q4)

You borrow HK$1,000,000 from a friend to help buy a flat. You have to pay the friend back HK$100,000 in the year 2016, and then pay back HK$100,000 in the year 2020, and then pay back HK$100,000 in the year 2024, and so on in a repeating pattern until you die. If you look at the years and their relationship to multiples of 4 you can work out a simple formula that uses modulus (%) to determine whether any year is a year where you have to pay back money.

A simple program that takes a year from the user and tells you whether it is a 'pay back money' year or not is shown below. You need to add the missing code in the _____ area shown below. You cannot alter any of the code given to you or add any new code. You must use the == given to you. Your answer must use modulus (%).
(When you use the program you only enter years greater than 2015).

```
year = int(input("Please enter a year: "))


if _____________________ == _____________________:
    print("In the year", year, "you have to pay back HK$100,000!!")
else:
    print("In the year", year, "you don't need to pay back any money")
```

Q5)

Here is a program which generates some music notes.

```
# <-- Here various modules are imported and MIDI is initialised
#     as usual; to save space the code is not shown

for instrument in range(48, 12, -12):
    output.set_instrument(instrument)

    for note in range(58, 78, 4):
        output.note_on(note, 127)
        time.sleep(0.5)
        output.note_off(note, 127)

# <-- At the end MIDI is stopped as usual; the code is not shown
```

How many notes are generated in total?
One note = a 'note on' message followed by a 'note off' message.

Answer: _____

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Q6)

An infinite loop is a loop that cannot end (unless you do *Ctrl-C* or turn off the computer!). For example, in the following programs the word 'finished' will not be printed if the program contains an infinite loop, because the program will never reach the last line of code. For each of the following programs, write 'YES' if the program has an infinite loop or 'NO' if it doesn't have an infinite loop. (3 marks each).

		Write YES if the program contains an infinite loop. Write NO if the program does not contain an infinite loop.

A	`while True:`
	`  print("1021!")`
	`print("finished")`	

B	`finished=True`
	`x=0`
	`while not finished:`
	`  print("I love you!")`
	`  if x > 5:`
	`  finished=True`
	`print("finished")`	

C	`countdown=2`
	`while countdown<9:`
	`  countdown=countdown-1`
	`  if countdown:`
	`  break`
	`  print("You love me!")`
	`print("finished")`	

D	`thisvar = 1`
	`while thisvar == 1:`
	`  num = input("Enter a number:")`
	`  print("You entered: ", num)`
	`print("finished")`	

E	`for x in range(33):`
	`  if x == 3:`
	`  continue`
	`  print("I am counting:", x)`
	`print("finished")`	

F	`mylist=["dog"]`
	`while len(mylist)>0:`
	`  if mylist[-1] == "dog":`
	`  mylist.append("cat")`
	`  elif mylist[0] == "dog":`
	`  mylist.remove("cat")`
	`  print(mylist)`
	`print("finished")`	

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Q7)

Here is a program which draws some shapes.

```
import turtle

turtle.reset()      # Use reset() to initialize the turtle
turtle.setup(800, 800) # Fix the width of the screen to 800 * 800

turtle.width(5)     # Set the width to 5 pixels
turtle.color("blue") # Set the line colour
turtle.speed(10)     # Draw faster
turtle.up()
turtle.goto(-300,-300)
turtle.down()
turtle.left(90)

for one_thing in range(4):
    turtle.forward(600)
    turtle.right(90)

    for another_thing in range(5):
        turtle.forward(100)
        turtle.right(144)
        turtle.dot(20)

turtle.done()
```

What shapes do you see on the screen? Fill in the missing blanks below.
(If, for example, you saw one hexagon on top of three squares and no other shapes, your answer would be 0 filled circles, 0 unfilled circles, 0 stars, 3 squares and 1 hexagon).

How many complete **filled circles** are drawn in total? Answer: _____

How many complete **unfilled circles** are drawn in total? Answer: _____

How many complete **stars** are drawn in total? Answer: _____

How many complete **squares** are drawn in total? Answer: _____

How many complete **hexagons** are drawn in total? Answer: _____

You must write an integer number (zero or higher) for each of the 5 answers.

Q8)

Show that you understand the following code by filling in the blanks underneath.

```
x="be more like sheep says cy"
part1=x[int(len(x)*.5):int(len(x)*.7)]
part2=x[19:22]
part3=x[0]+x[-6]+x[-6]
space=" "
result=part1+space+part2+space+part3+space+part3

print(result)
```

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You don't have to use speech marks for any of the answers, but you can if you want to e.g. `"cat"`.

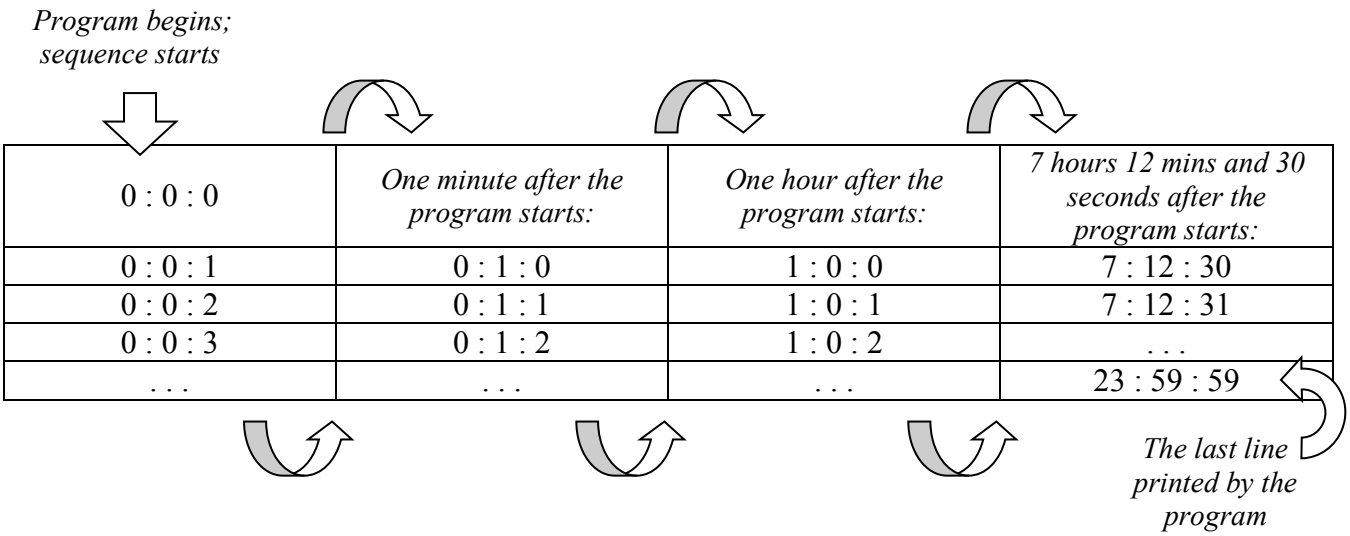
`part1` is _____

`part2` is _____

`part3` is _____

`result` is _____

Q9)
A program has been written which acts like a clock/stopwatch. Every second, the clock is displayed by the program. The program output sequence is indicated below:



As you can see, the first line printed by the program is 0 : 0 : 0 and the last line printed by the program is 23 : 59 : 59.

The code is shown below. As you can see, three lines of code are missing. The appropriate indentation is already given to you. Write the missing lines of code in the appropriate places. You can only write one line of code in each place. You cannot change any of the code shown. Your code must be clear and correct (i.e. it would work if you really ran your code) to get the marks.

```
import time

for hours in range(24):

    _____________________________________________________________________________

        _____________________________________________________________________________

            print(hours, ":", minutes, ":", seconds)

            _____________________________________ # Delay for 1 second
```

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Q10)
This program…

```
import turtle

colour_list=["black","grey10","grey20","grey30", \
             "grey40","grey50","grey60","grey70", \
             "grey80","grey90","white"]
turtle.up()
turtle.goto(-400,100)
turtle.down()

for name in colour_list: # Go through every colour, put it in 'name'
    turtle.fillcolor(name) # Use the colour to fill the square
    turtle.begin_fill() # Start filling

    for _ in range(4): # Draw a square
        turtle.forward(70)
        turtle.left(90)

    turtle.end_fill() # Stop filling

    turtle.forward(70) # Move to the right, ready for the next square

turtle.done() # Finished
```

… produces this result:



The list of colours in the next program is exactly the same as the list of colours in the above program. What does the following program produce?

```
import turtle

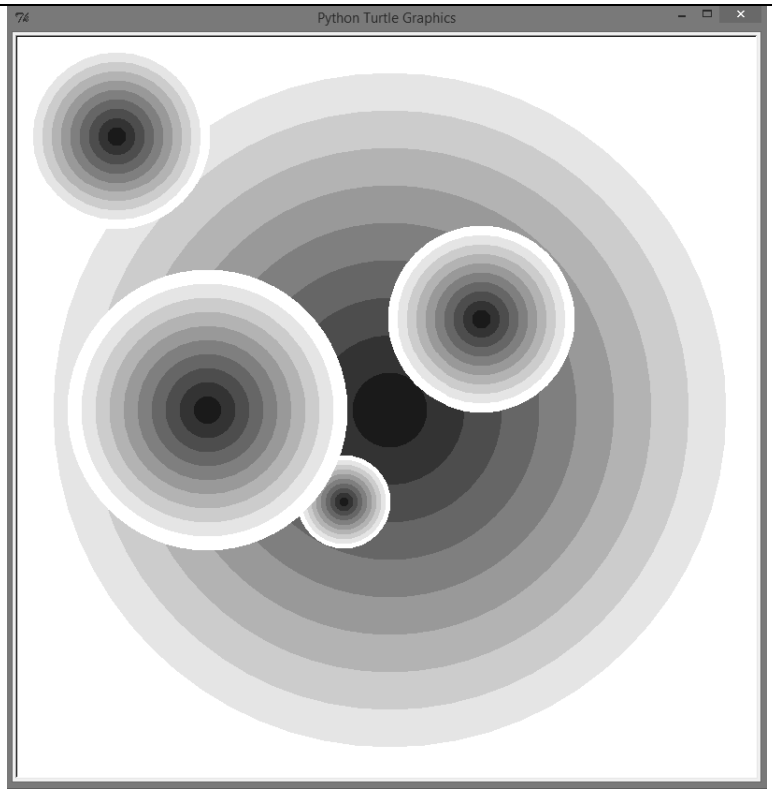
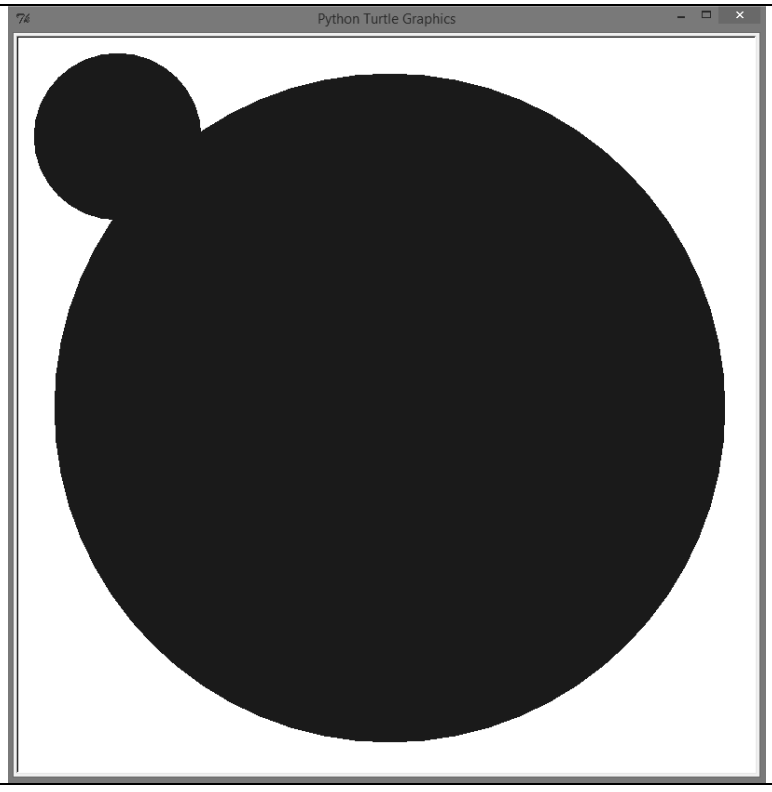
def apply_effect(max_radius):
    colour_list=["black","grey10","grey20","grey30", \
                 "grey40","grey50","grey60","grey70", \
                 "grey80","grey90","white"]
    index=len(colour_list)-1

    for size in range(max_radius, 0, -1*(max_radius//10) ):
        turtle.color(colour_list[index])
        index=index-1
        turtle.dot(size)

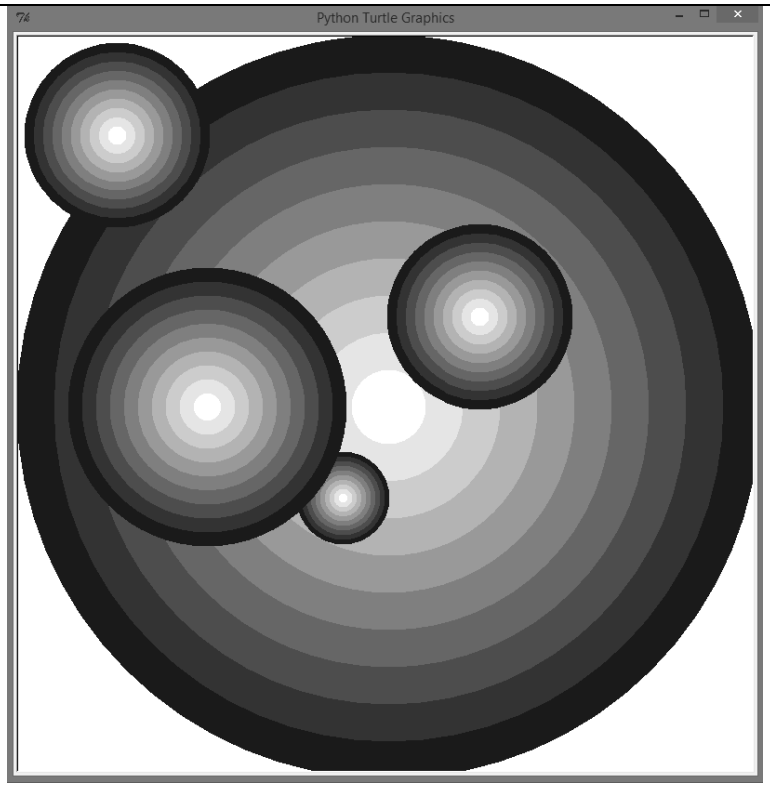
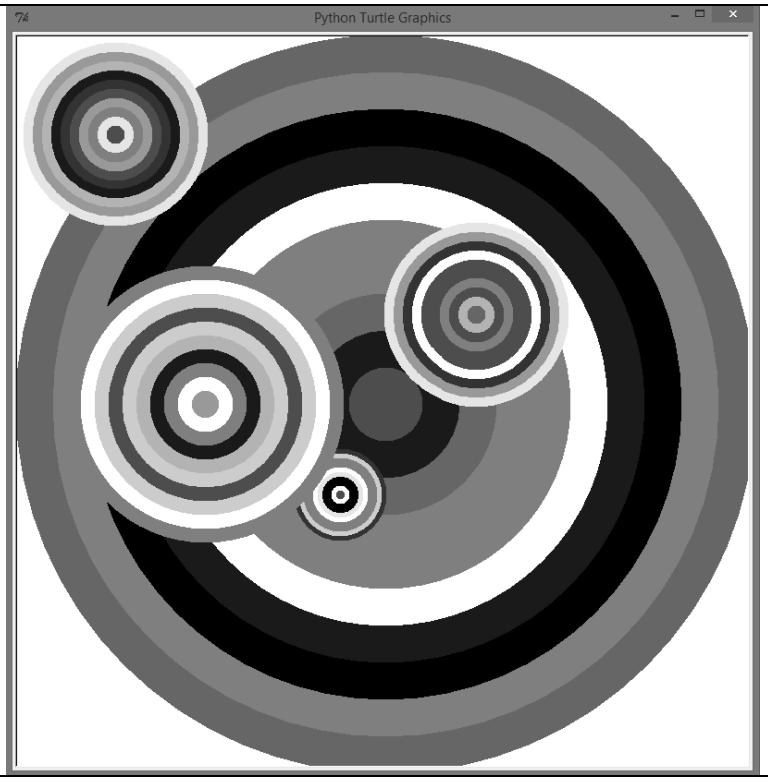
turtle.setup(800, 800) # Fix the width of the screen to roughly 800 * 800
turtle.up()
apply_effect(800)
turtle.goto(-300,300)
apply_effect(200)
turtle.goto(-50,-100)
apply_effect(100)
turtle.goto(100,100)
apply_effect(200)
turtle.goto(-200,-0)
apply_effect(300)
turtle.done()
```

Choose one of A/B/C/D/E shown on the following pages. Answer (A/B/C/D/E) : _____

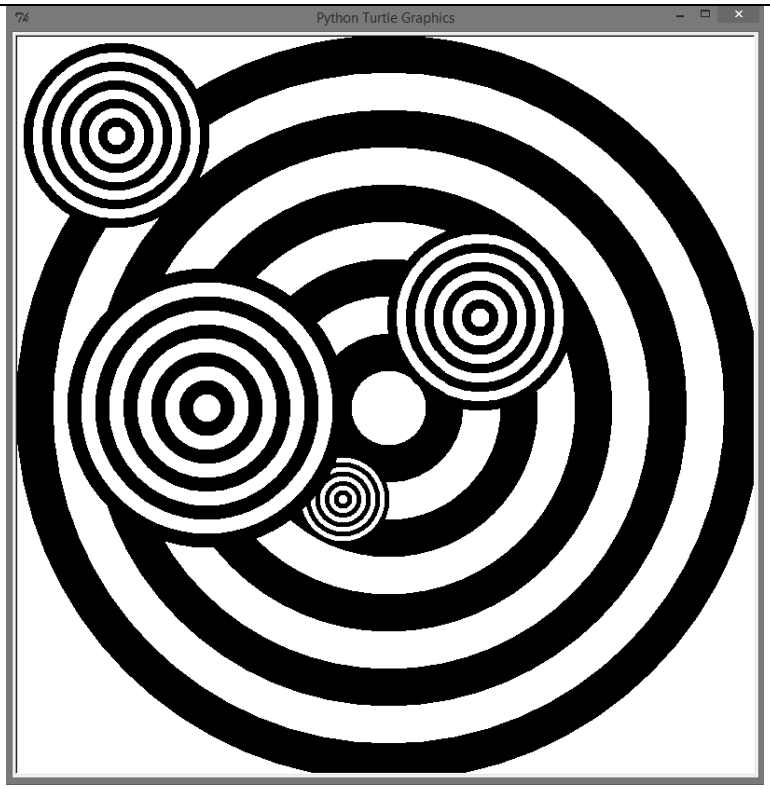
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A	
B	

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C	
D	

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E	

- End of the exam -